

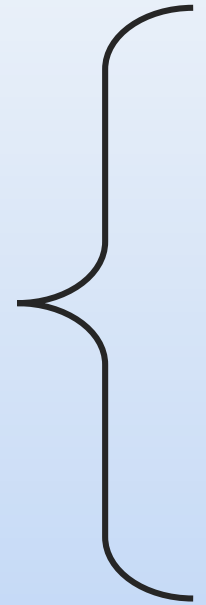
Hotel Reservation Management System

20042134 - Vinicius Quintanilha Sano

Module: B8IT150 – Advanced Programming

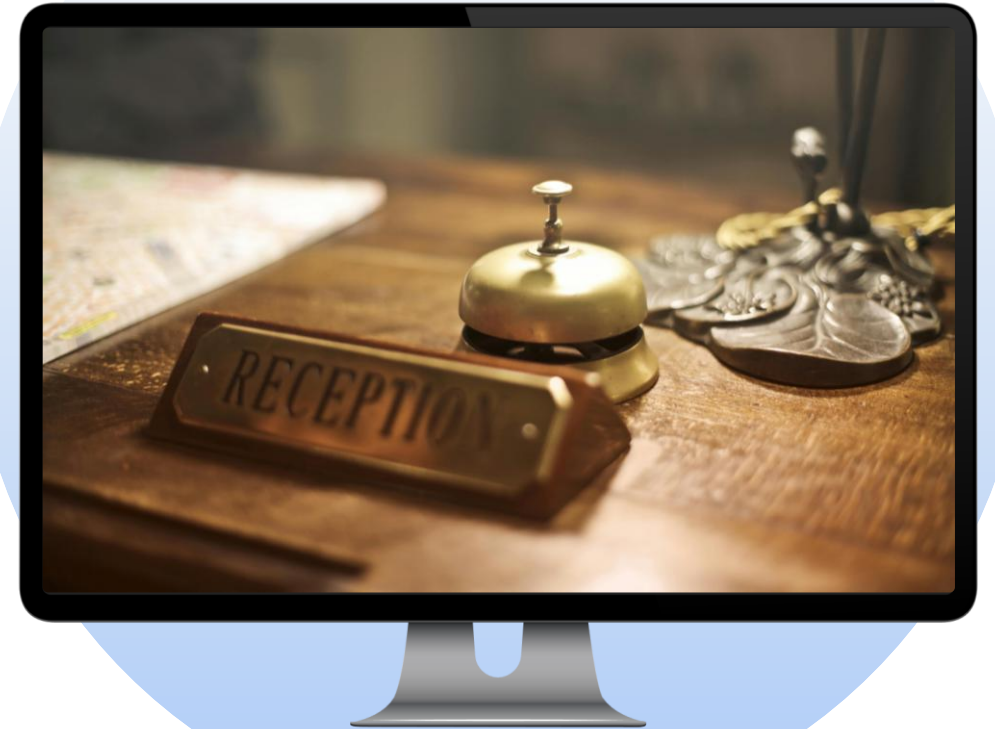
Lecturer: Kanza Ali Manzar

Agenda

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- Developer's Background
 - Project's Background
 - UML Diagram
 - ERD Diagram
 - Features and Functionalities
 - Future Implementations

Developer's Background

- 4 years experience in the hospitality industry
- Use hotel's management system daily
- No professional experience with software development or any related IT area



4 Project's Background

Purpose: To provide hotels with an efficient system for managing, tracking, and organizing guest reservations

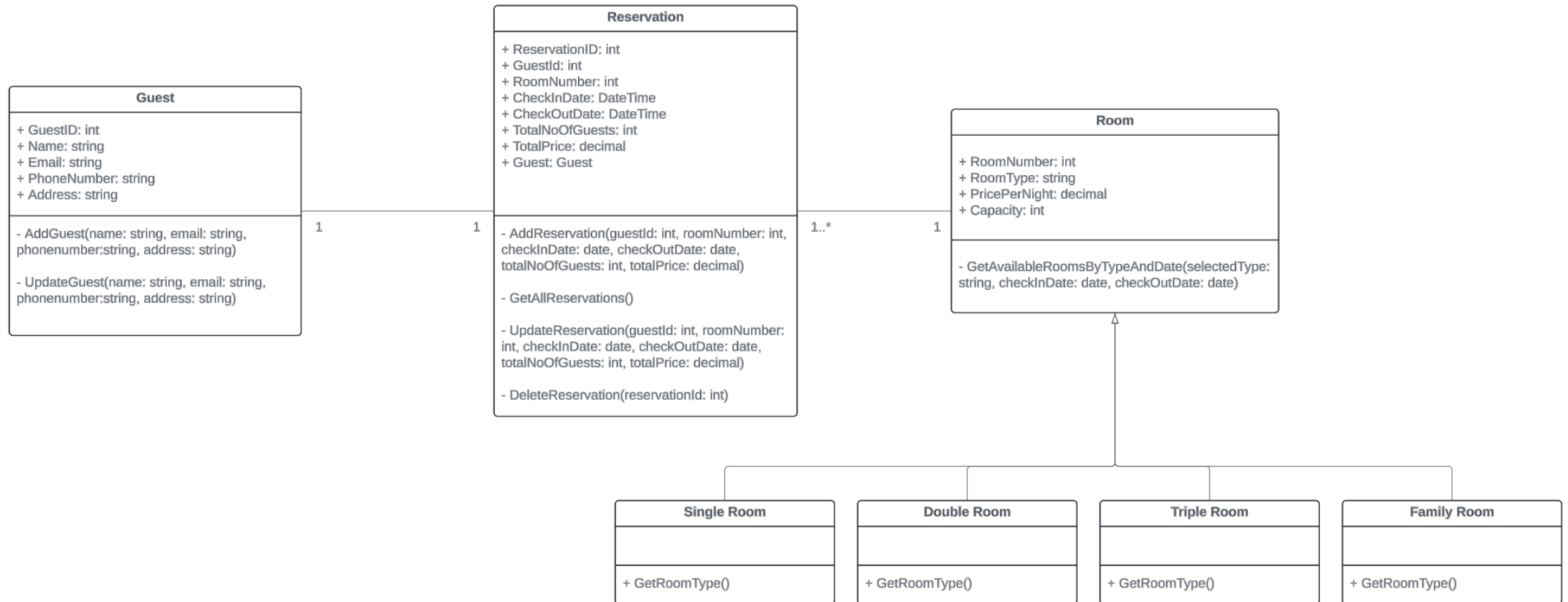
- ✓ By creating, organizing and storing guest's information (*name, email, phone number, address*)
- ✓ By creating, organizing and storing reservations (*check-in and check-out date, room type and number, rate, total n° of guests*)
- ✓ By allowing quick access and modification of guest's information and reservations



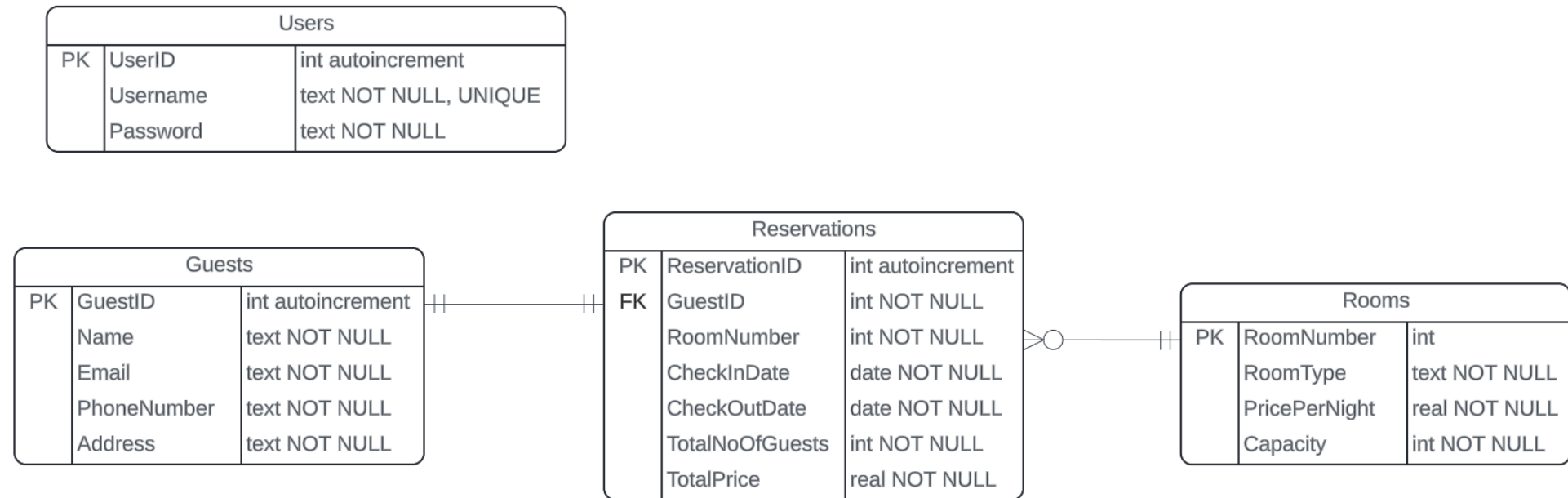
The screenshot shows a web form titled "Hotel Reservation System". It contains the following fields and controls:

- Guest Name:** A text input field with a placeholder "Enter Guest's Name".
- Email:** A text input field.
- Phone Number:** A text input field.
- Address:** A text input field.
- Total number of guests:** Radio buttons for 1, 2, 3, and 4. The "1" option is selected.
- Check-in Date:** A date picker showing "Thursday 4 December 2025".
- Check-out Date:** A date picker showing "Thursday 4 December 2025".
- Select room type:** A dropdown menu with "Choose Room" selected.
- Select Room Number:** A text input field.
- Total Rate:** A text input field.
- Create Reservation:** A button at the bottom.

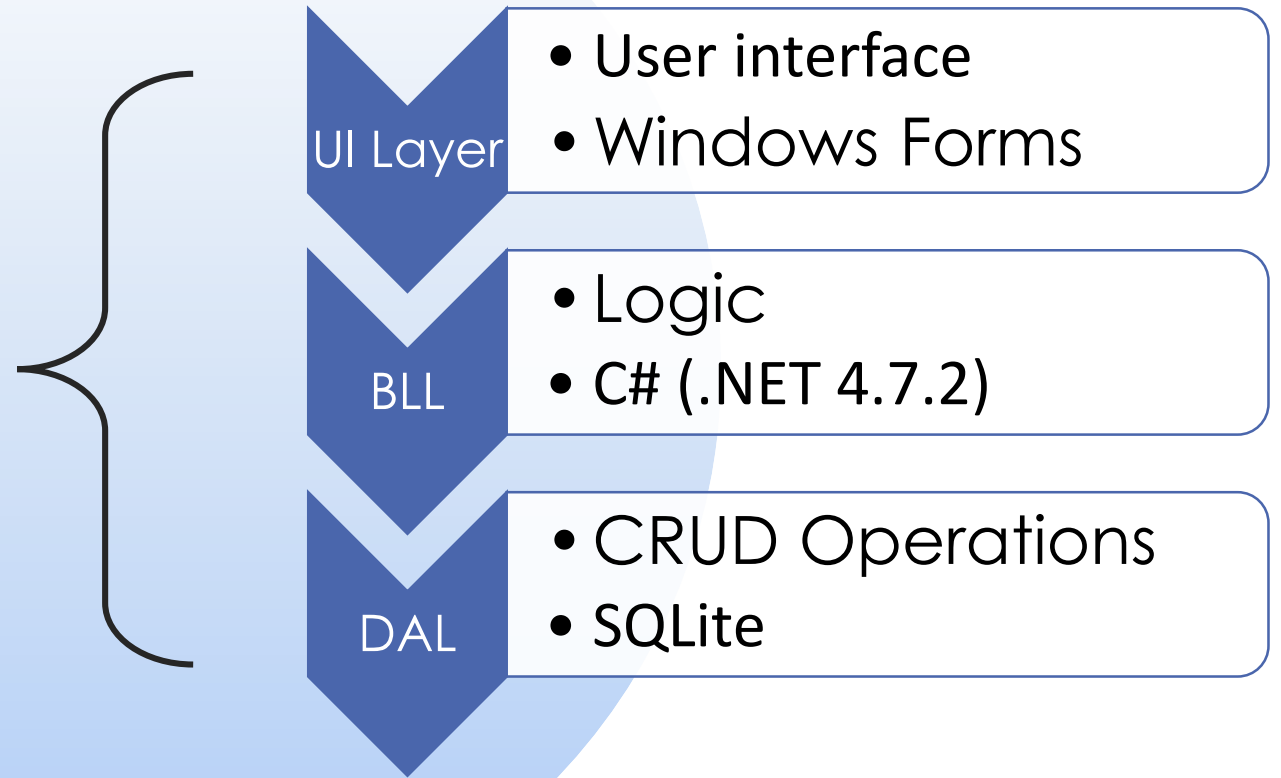
UML Diagram



ERD Diagram



Features and Functionalities

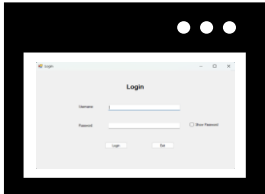


Features and Functionalities

Presentation Layer (UI Layer)

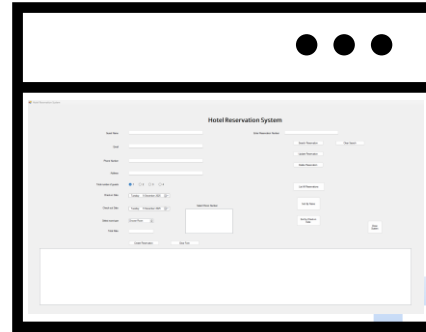
- Windows Forms App (.NET Framework v. 4.7.2)

Login Form



- I. Text boxes to:
 - A. Input user's credentials (username and password)
- II. Buttons to:
 - A. Login
 - B. Exit application
- III. Check box to:
 - A. Show or hide password

Hotel Reservation System Form



- I. Text boxes to:
 - A. Input guest's information and reservation details
- II. Buttons to:
 - A. Create reservation
 - B. Search for specific reservation
 - C. List all reservations
 - D. Sort all reservations
 - E. Update and delete reservation
 - F. Exit application
- III. List boxes to:
 - A. Show available rooms
 - B. Show Reservations

Features and Functionalities

Business Logic Layer (BLL)

Classes:

Guest

Reservation

Room

Encapsulation

```

9 references
public class Guest
{
    7 references
    public int GuestID { get; set; }
    10 references
    public string Name { get; set; }
    6 references
    public string Email { get; set; }
    6 references
    public string PhoneNumber { get; set; }
    6 references
    public string Address { get; set; }

    1 reference
    public Guest(string name, string email, string phoneNumber, string address)
    {
        Name = name;
        Email = email;
        PhoneNumber = phoneNumber;
        Address = address;
    }
}

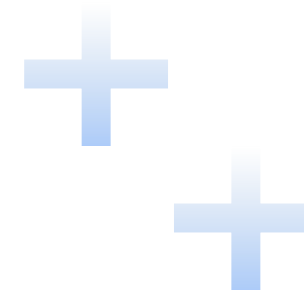
```

Single

Double

Triple

Family



Abstraction

```

21 references
public abstract class Room
{
    5 references
    public int RoomNumber { get; set; }
    5 references
    public decimal PricePerNight { get; set; }
    1 reference
    public bool IsAvailable { get; set; }
    5 references
    public int Capacity { get; set; }

    4 references
    protected Room(int roomNumber, decimal pricePerNight, bool isAvailable, int capacity)
    {
        RoomNumber = roomNumber;
        PricePerNight = pricePerNight;
        IsAvailable = isAvailable;
        Capacity = capacity;
    }

    5 references
    public abstract string GetRoomType();
}

```

Inheritance

```

2 references
public class SingleRoom : Room
{
    1 reference
    public SingleRoom(int number, decimal price, bool available, int capacity)
        : base(number, price, available, capacity)
    {
    }

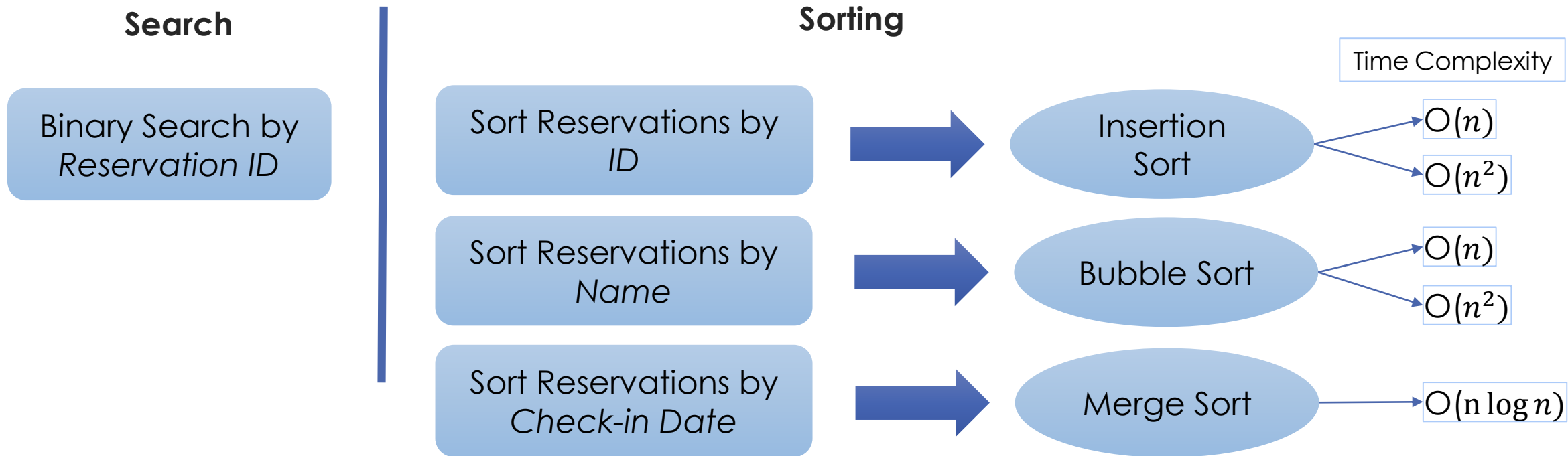
    2 references
    public override string GetRoomType()
    {
        return "Single Room";
    }
}

```

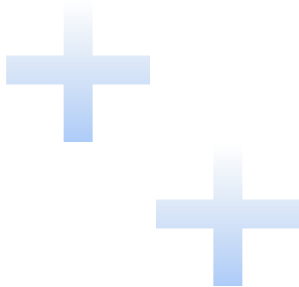
Features and Functionalities

Business Logic Layer (BLL)

Services:



Features and Functionalities



Business Logic Layer (BLL)

Validation:

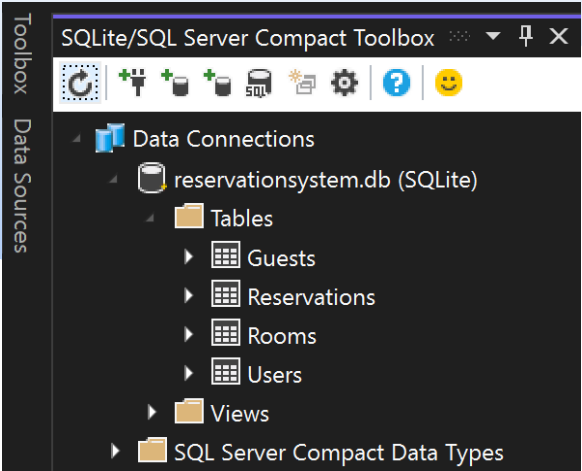
Guest's Info	Name	Email	Phone Number	Address
Rules	At least first and last name with minimum of 3 letters each.	Must contain characters before and after the @ symbol, a valid domain, and at least 2 characters after the dot.	Must include a country code (1–3 digits) followed by a space and the phone number (8–15 digits).	Cannot be empty and may contain letters, numbers, spaces, commas, and hyphens.

Reservation's Details	Capacity Room	Check-in Check-out Date	Room Type and Number	Total Price
Rules	Cannot exceed the room's maximum capacity.	1) Check-out date must be after check-in date. 2) New reservations must be for the present or future; past dates are not allowed.	A room must be selected.	Must be a valid positive decimal number.

12 Features and Functionalities

Data Access Layer (DAL)

SQLite / SQL Server Compact Toolbox

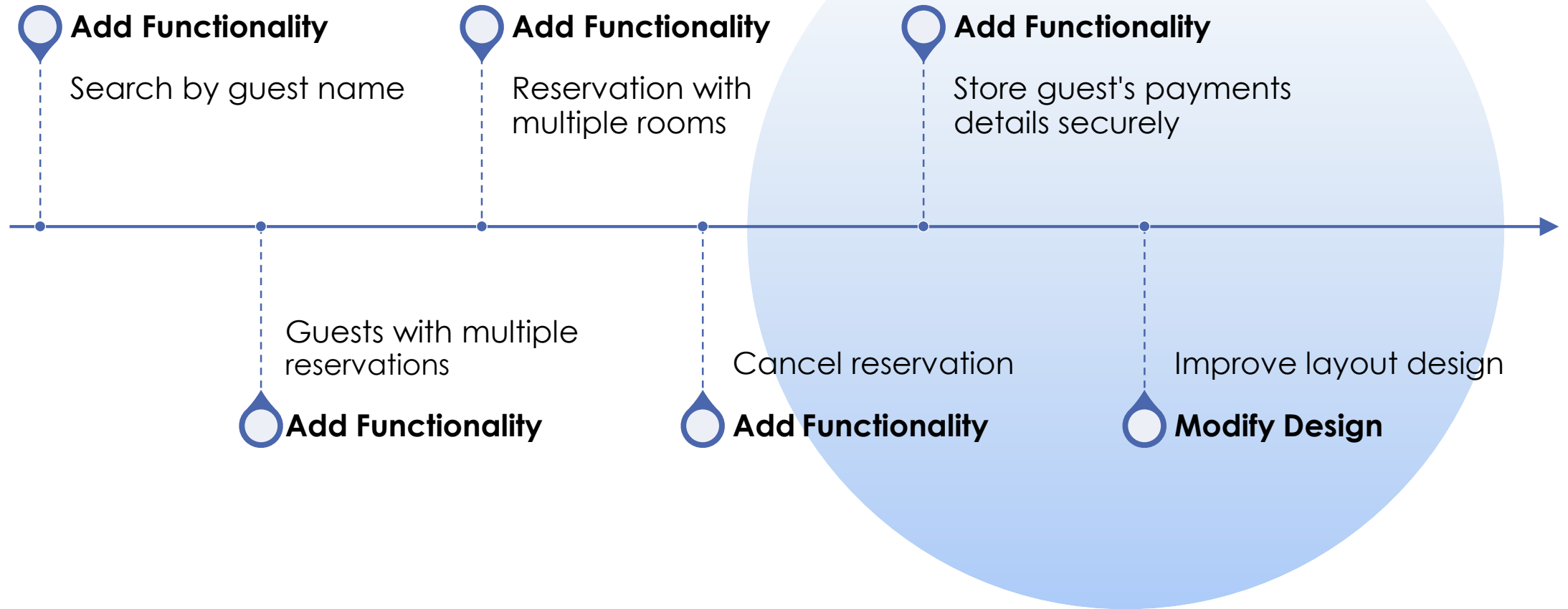


- I. Create SQL Database (reservationsystem.db)
- II. Create table with guest's information
- III. Create table with reservation details
- IV. Create table with rooms details
- V. Retrieve information about guests and reservations
- VI. Update Information about guests and reservations
- VII. Delete reservation

CRUD Operations

Create	Read	Update	Delete
Guests	Guest's info	Guest's info	Reservations
Reservations	Reservations' details	Reservations' details	
Rooms	Rooms' details		
Users	Users' credentials		

13 Future Implementations



Thank you

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